ML BASED SMART PRODUCTIVITY TRACKER

PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this project report titled **“ML BASED SMART PRODUCTIVITY TRACKER”** is the bonafide work of “2116230701284-K SANJAKUMARAN”, ”2116230701323-N S SMITH”, “2116230701329-R SOWNDARIYA” the student carried out under the supervision of the faculty guide.

SIGNATURE OF THE FACULTY INCHARGE

Submitted for the Practical Examination held on ____________

SIGNATURE OF THE INTERNAL EXAMINER

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I. ABSTRACT

The ML Based Smart Productivity Tracker is an intelligent desktop application designed to monitor, analyze, and improve user productivity using machine learning techniques. Traditional productivity tracking systems only provide static reports and lack intelligent analysis of user behavior. This proposed system addresses these limitations by integrating real-time activity monitoring, data analytics, and AI-powered productivity prediction.

The system continuously tracks user activities such as active applications, keystrokes, mouse clicks, idle time, and session duration. These activities are stored in an SQLite database and analyzed using a machine learning model based on the XGBoost algorithm. The trained model classifies activities into Productive, Neutral, or Unproductive categories and generates a productivity score between 0 and 100.

A modern graphical dashboard built with Tkinter and Matplotlib provides real-time visualization of productivity statistics, weekly reports, activity trends, and machine learning analysis. The system also generates charts for category-wise usage, daily productivity trends, and feature importance analysis. By combining intelligent analytics with interactive visualization, the application helps users understand work patterns, minimize distractions, and improve overall efficiency.

The ML Based Smart Productivity Tracker demonstrates how AI and behavioral analytics can be applied to personal productivity management, offering a scalable and efficient solution for students, professionals, and organizations.

II. LITERATURE REVIEW

Several research studies have explored productivity monitoring, behavioral analytics, and machine learning-based classification systems. Existing productivity applications mainly focus on time tracking but lack intelligent decision-making and adaptive recommendations.

Recent advancements in machine learning and activity recognition have enabled systems to classify user behavior based on application usage patterns, keyboard activity, and session duration. Ensemble learning algorithms such as Random Forest and XGBoost have shown high accuracy in predictive analytics and productivity classification tasks.

Studies on behavioral computing highlight that integrating real-time monitoring with AI-based analysis can significantly improve focus management, reduce distractions, and optimize workflow efficiency. Dashboard visualization systems using Python libraries such as Matplotlib and Tkinter have also been widely adopted for presenting analytics in an interactive and understandable manner.

The proposed Smart Productivity Tracker builds upon these advancements by integrating real-time monitoring, machine learning prediction, visualization dashboards, and historical analytics into a single desktop application.

III. PROPOSED SYSTEM

The proposed system is an AI-powered productivity tracking framework that continuously monitors user activities and predicts productivity levels using machine learning algorithms.

Objectives:
1. To monitor real-time system activity.
2. To classify activities as productive, neutral, or unproductive.
3. To generate intelligent productivity scores using machine learning.
4. To provide graphical visualization of activity trends.
5. To maintain historical productivity reports using SQLite.

Features:
- Real-time activity tracking
- SQLite database integration
- XGBoost machine learning model
- Interactive dashboard
- Weekly productivity analytics
- Feature importance analysis
- Productivity scoring system

System Architecture Modules:
1. Activity Monitoring Module
2. Database Management Module
3. Machine Learning Engine
4. Dashboard Visualization Module
5. Report Generation Module

IV. MODULES DESCRIPTION

4.1 Activity Monitoring Module

This module collects real-time information about active windows, keystrokes, mouse clicks, session duration, and idle time. The collected data are processed and prepared for storage and analysis.

4.2 Database Management Module

The database module uses SQLite to store activity sessions, productivity labels, timestamps, and prediction results. It ensures efficient retrieval and management of historical data.

4.3 Machine Learning Module

The ML module uses the XGBoost classification algorithm to predict productivity categories. The model is trained using activity data and generates productivity scores based on user behavior.

4.4 Dashboard Visualization Module

This module provides a graphical user interface built with Tkinter and Matplotlib. It displays live productivity scores, activity charts, weekly reports, and category analysis.

4.5 Report Generation Module

The reporting module generates productivity summaries, charts, and historical analytics to help users evaluate their performance over time.

V. IMPLEMENTATION AND RESULTS

The system was implemented using Python as the primary programming language. The Tkinter framework was used for GUI development, SQLite for database management, and XGBoost for machine learning analysis.

Hardware Requirements:
- Intel Core i5 Processor or above
- 8 GB RAM
- 500 GB Storage

Software Requirements:
- Python 3.x
- Tkinter
- SQLite
- Scikit-learn
- XGBoost
- Pandas
- Matplotlib

Implementation Steps:
1. Real-time activity collection
2. Database storage using SQLite
3. Feature engineering and preprocessing
4. XGBoost model training
5. Dashboard integration
6. Visualization and reporting

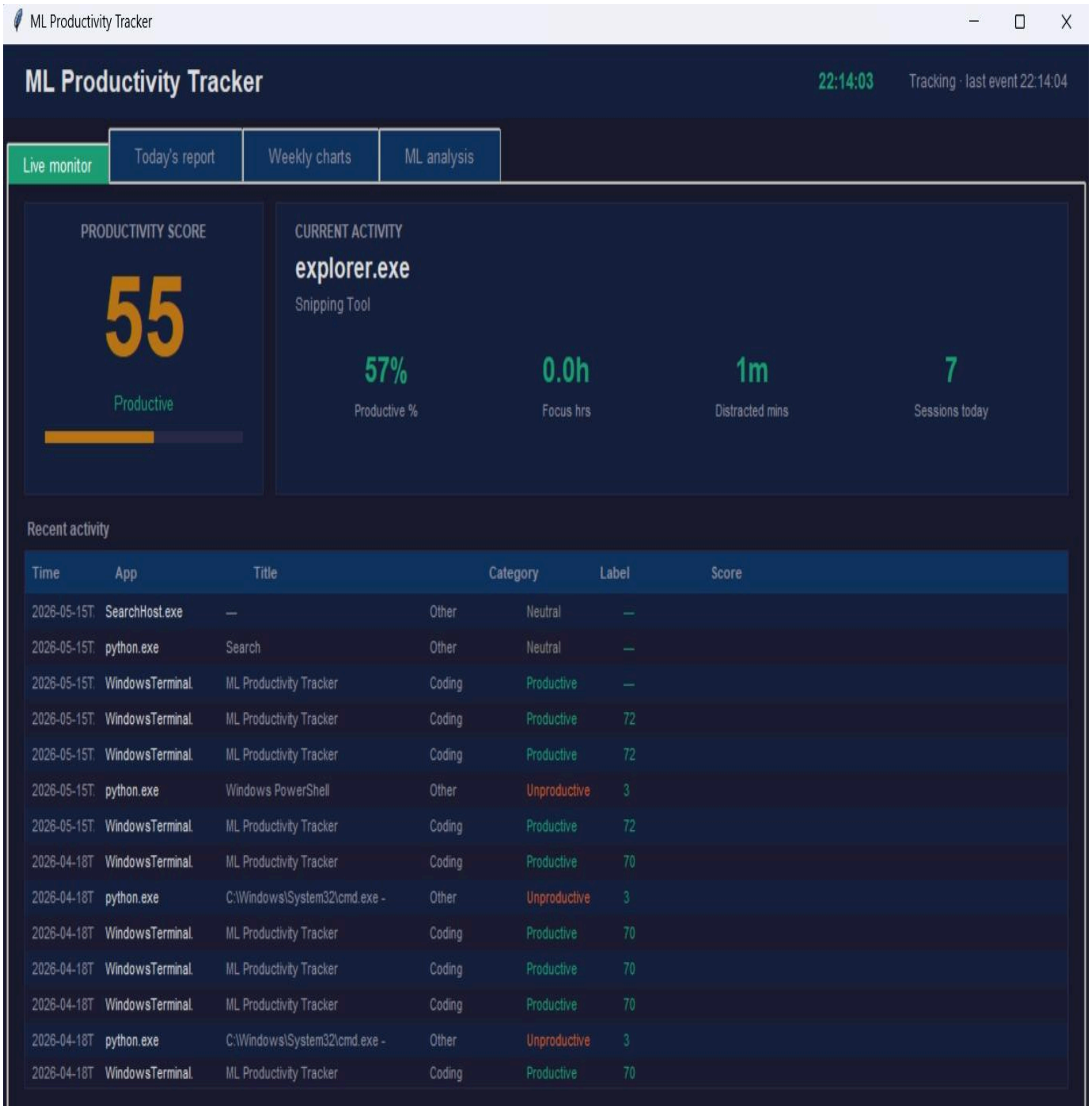
Results:
The application successfully tracked user activities and generated productivity scores with high accuracy. The ML model effectively classified productive and unproductive behavior patterns, helping users identify distractions and improve work efficiency.

VI. CONCLUSION AND FUTURE WORK

The ML Based Smart Productivity Tracker successfully demonstrates the integration of machine learning and real-time activity monitoring for productivity analysis. The system provides accurate productivity classification, graphical insights, and intelligent analytics that help users improve focus and efficiency.

Future enhancements may include:

- Cloud synchronization
- Mobile application support
- Deep learning-based analytics
- AI-based productivity recommendations
- Multi-user support
- Browser extension integration


ML Productivity Tracker
ML Productivity Tracker
22:14:03
Tracking · last event 22:14:04
Live monitor
Today's report
Weekly charts
ML analysis
PRODUCTIVITY SCORE
55
Productive
CURRENT ACTIVITY
explorer.exe
Snipping Tool
57%
Productive %
0.0h
Focus hrs
1m
Distracted mins
7
Sessions today
Recent activity
Time App Title Category Label Score
2026-05-15T SearchHost.exe — Other Neutral —
2026-05-15T python.exe Search Other Neutral —
2026-05-15T WindowsTerminal. ML Productivity Tracker Coding Productive —
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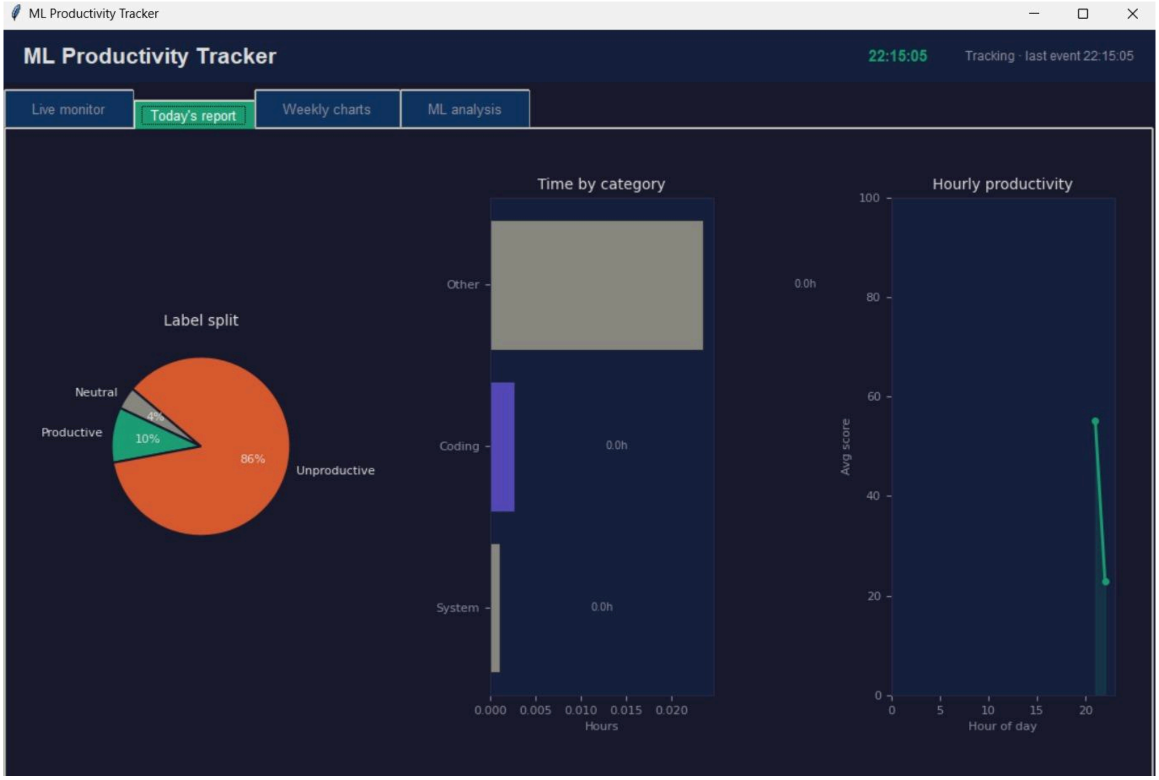
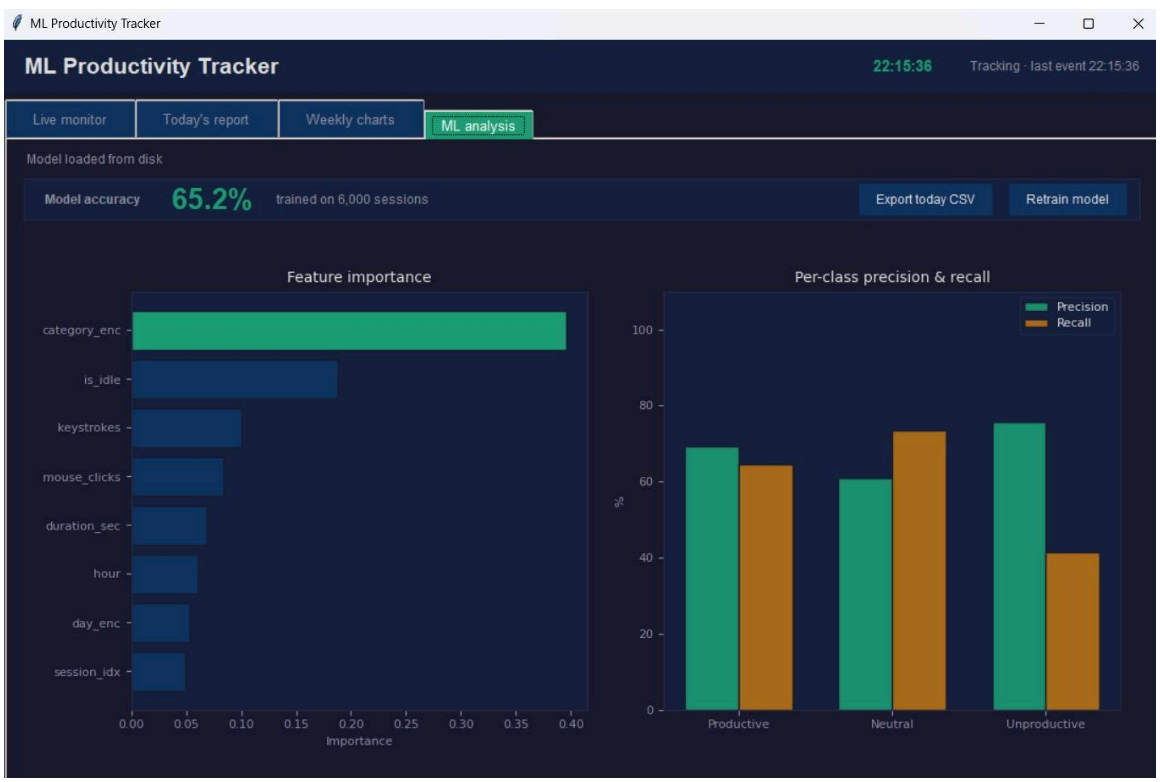
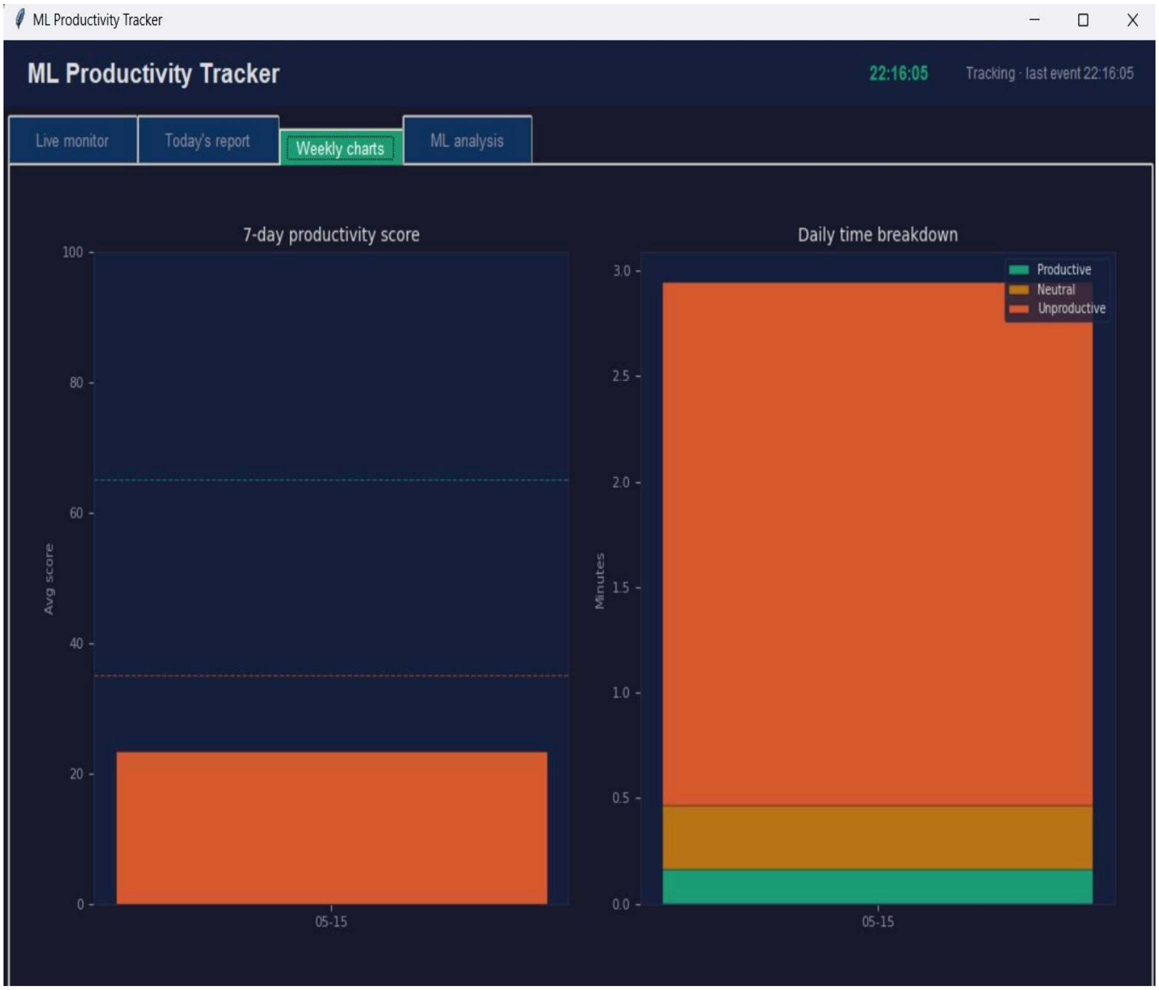


ML Productivity Tracker
ML Productivity Tracker
22:15:05
Tracking · last event 22:15:05
Live monitor
Today's report
Weekly charts
ML analysis
Label split
Neutral
Productive
Unproductive
4%
10%
86%
Time by category
Other
Coding
System
0.0h
0.0h
0.0h
0.000 0.005 0.010 0.015 0.020
Hours
Hourly productivity
Avg score
Hour of day



ML Productivity Tracker
ML Productivity Tracker
22:15:36
Tracking · last event 22:15:36
Live monitor
Today's report
Weekly charts
ML analysis
Model loaded from disk
Model accuracy
65.2%
trained on 6,000 sessions
Export today CSV
Retrain model
Feature importance
category_enc
is_idle
keystrokes
mouse_clicks
duration_sec
hour
day_enc
session_idx
Importance
Per-class precision & recall
Precision
Recall
%
Productive
Neutral
Unproductive



ML Productivity Tracker
ML Productivity Tracker
22:16:05
Tracking · last event 22:16:05
Live monitor
Today's report
Weekly charts
ML analysis
7-day productivity score
Avg score
100
80
60
40
20
0
05-15
Daily time breakdown
Productive
Neutral
Unproductive
Minutes
3.0
2.5
2.0
1.5
1.0
0.5
0.0
05-15

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